

Dark matter

Using gravity to investigate unseen mass

WVU DSPIRA astronomy

Adapted from lecture material credited to Mark Voit

How can we detect unseen mass?

Compare mass inferred from gravity with the matter we observe.

Use galaxy rotation, cluster motions, hot gas, and gravitational lensing.



Galaxy cluster image retained from the source lecture.

Explore orbits and mass

1. Open Gravity and Orbits and choose Model.
2. Change the central mass and orbital distance.
3. Show velocity and describe how the orbit changes.

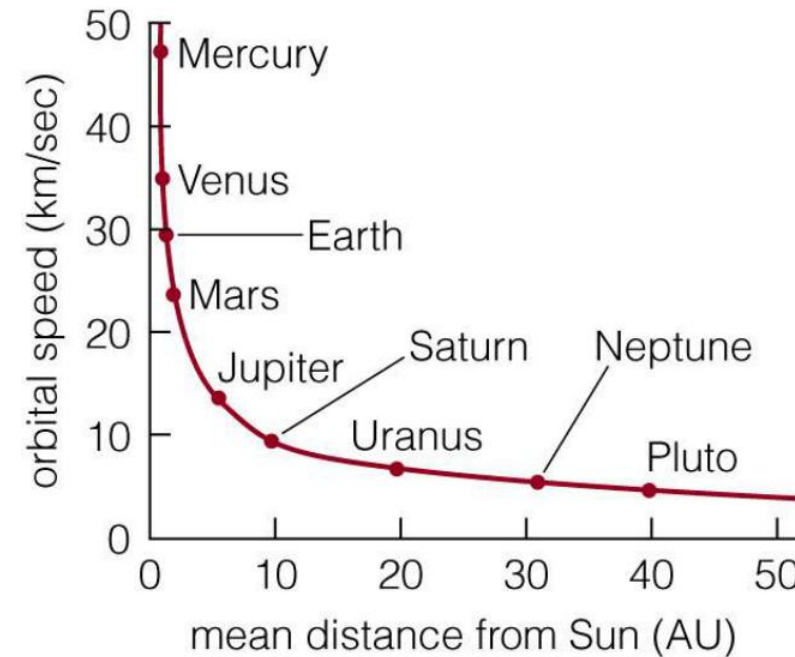
[Open Gravity and Orbits — PhET](#)

The Solar System rotation curve

The Sun contains nearly all the Solar System's mass.

For circular orbits, $v^2 = GM/r$.

With fixed central mass, orbital speed falls as radius increases.



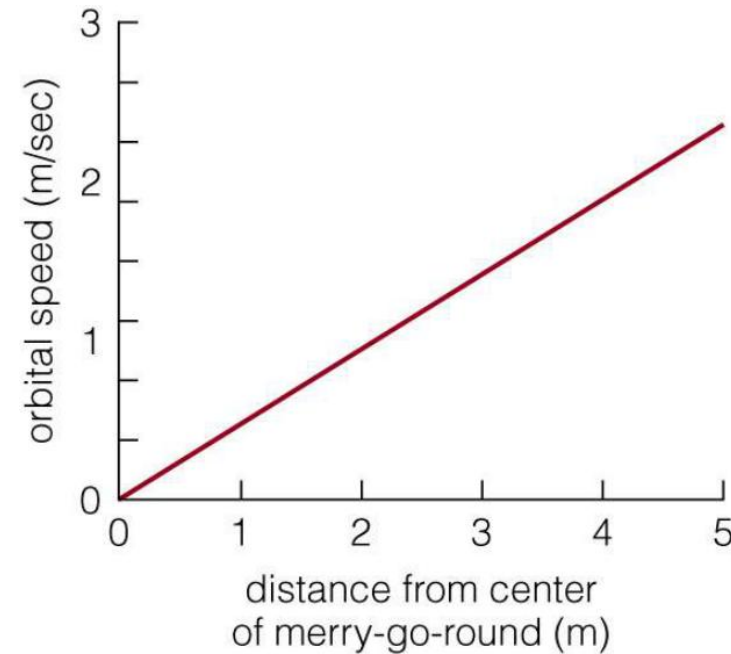
Orbital speed decreases with distance from the Sun.

Rotation of a merry-go-round

Every point completes a turn in the same time.

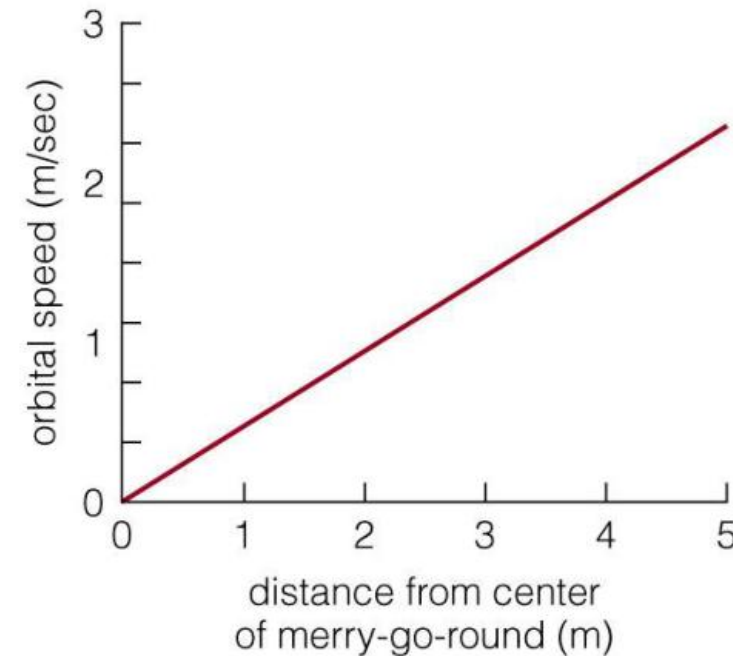
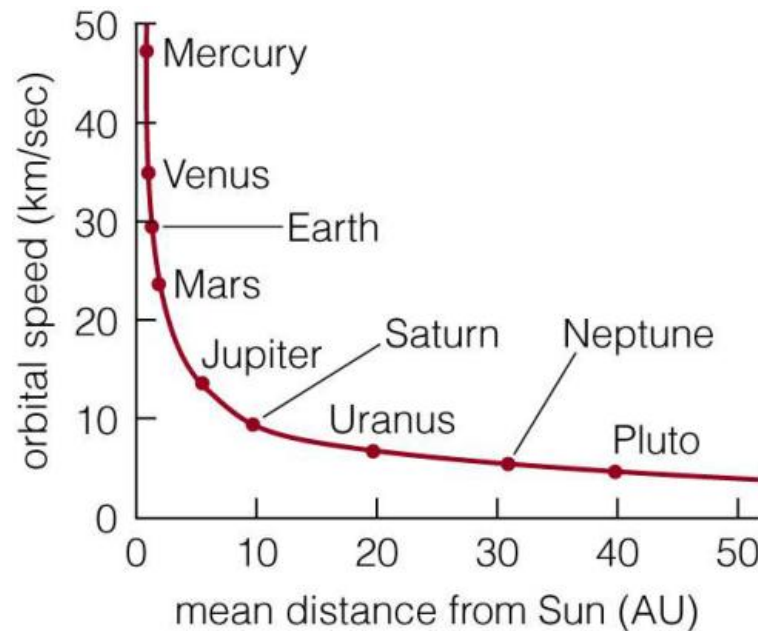
Angular speed ω is constant, so $v = \omega r$.

Points farther from the center move faster.



Rigid rotation gives a rising speed-versus-radius curve.

Predict the Galactic rotation curve



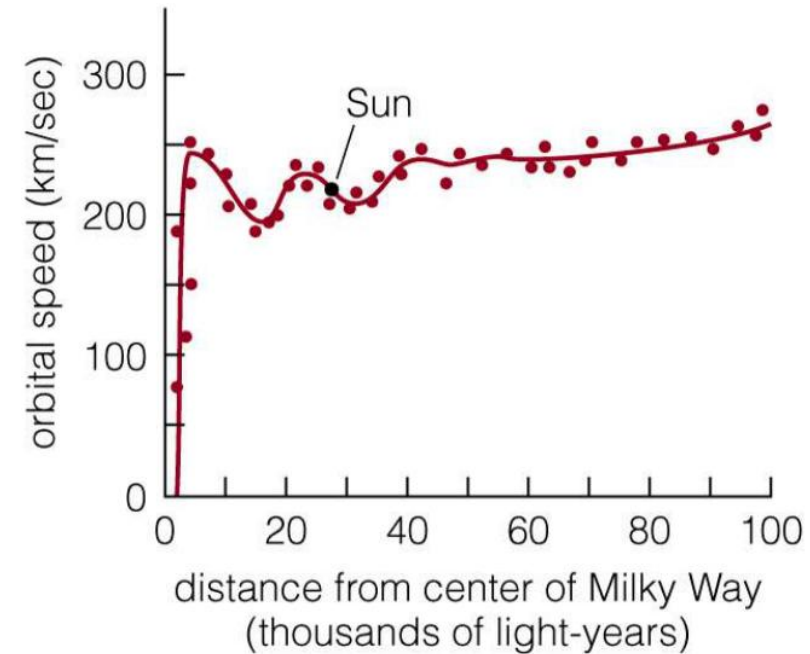
Will it fall like the Solar System, rise like rigid rotation, or follow another pattern? Explain your choice.

The Milky Way rotation curve

Outer orbital speeds change little with radius.

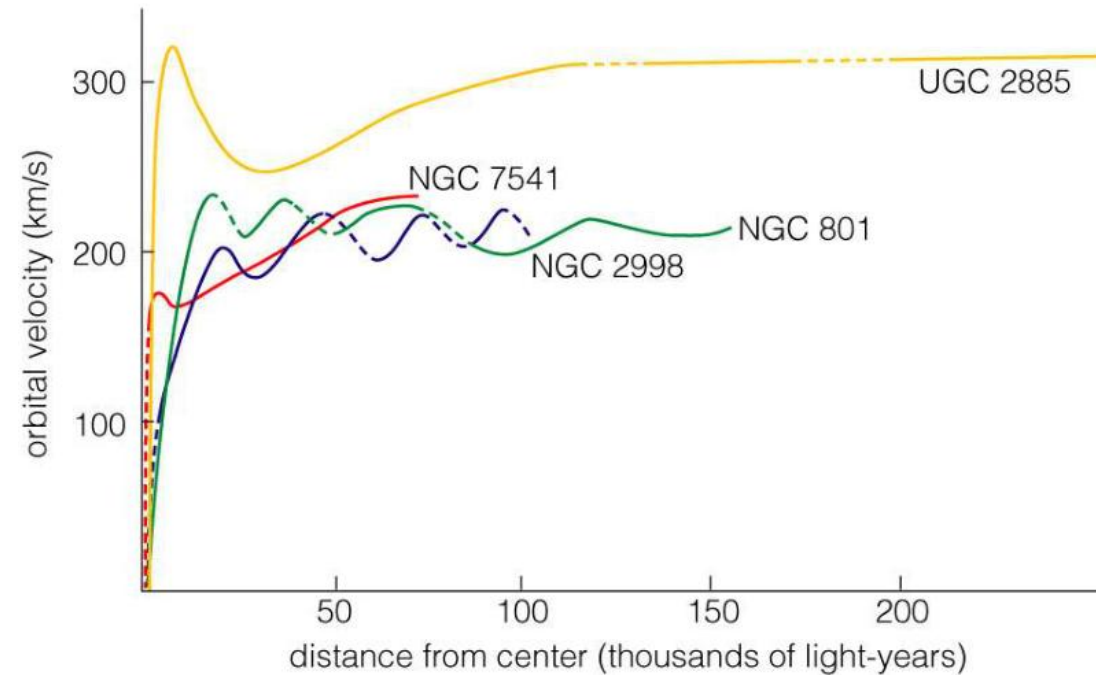
A concentrated central mass predicts a declining curve.

What mass distribution could explain the difference?



The outer rotation curve is approximately flat.

Other spiral galaxies show the pattern



Many spiral galaxies have extended, nearly flat rotation curves.

Compare two mass estimates

Matter we observe

Estimate stars and gas from their radiation.

Use a model to predict their gravitational effects.

Mass inferred from gravity

Measure orbital motions.

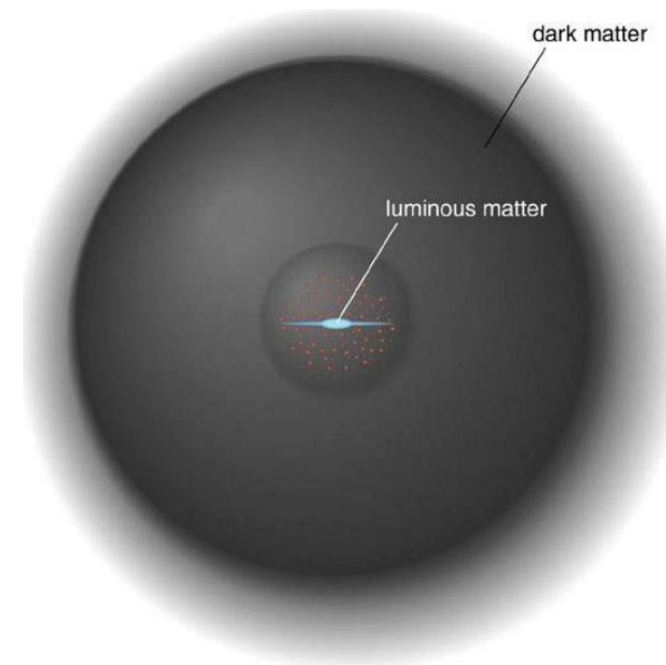
Compare the required mass with the observed stars and gas.

A dark matter halo

An extended halo adds mass beyond the bright disk.

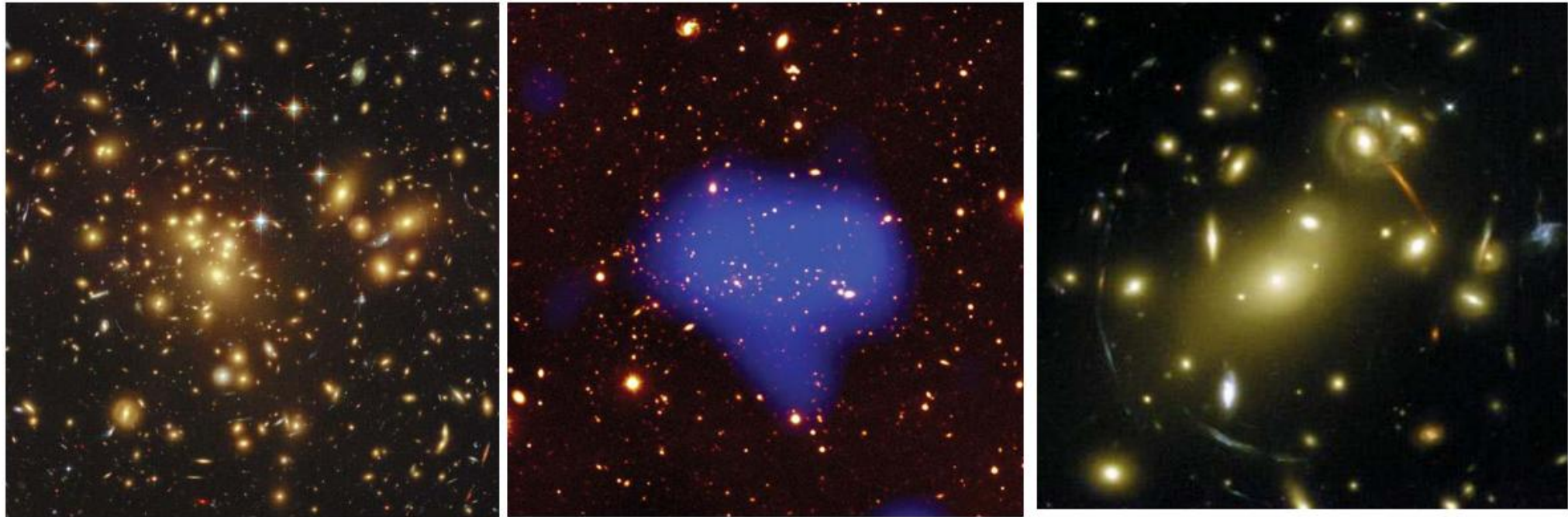
Its gravity can help explain the rotation curve.

The halo is inferred from observations and models.



The visible galaxy lies within a much larger halo.

Test the idea in galaxy clusters



Three probes: galaxy motions, hot gas, and gravitational lensing.

Measure galaxy motions

Measure the velocities of galaxies in a cluster.

Use their motions to estimate the mass needed to bind the system.

Stars alone do not supply enough mass.



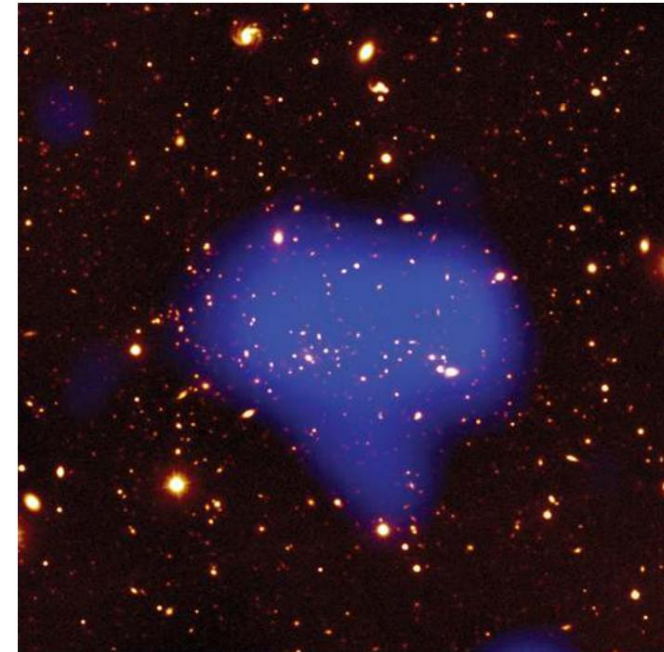
Galaxy motions probe the gravitational field of a cluster.

Measure hot gas with X-rays

Hot gas outweighs the stars.

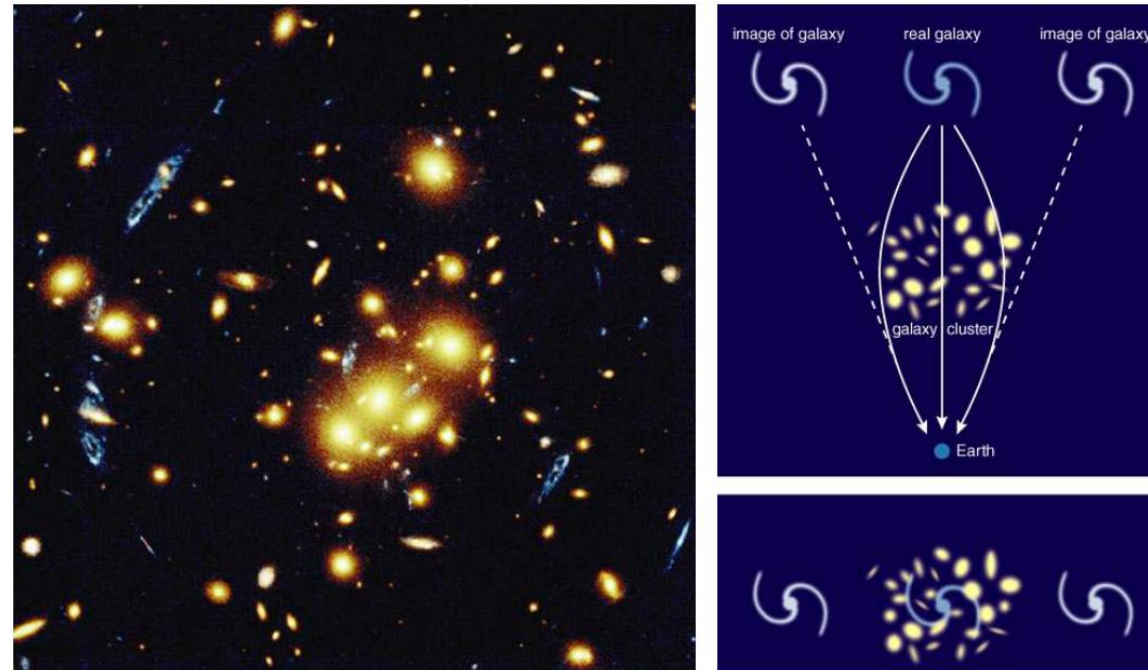
Gas temperature and density help us estimate the total mass.

That mass exceeds the gas and stars we observe.



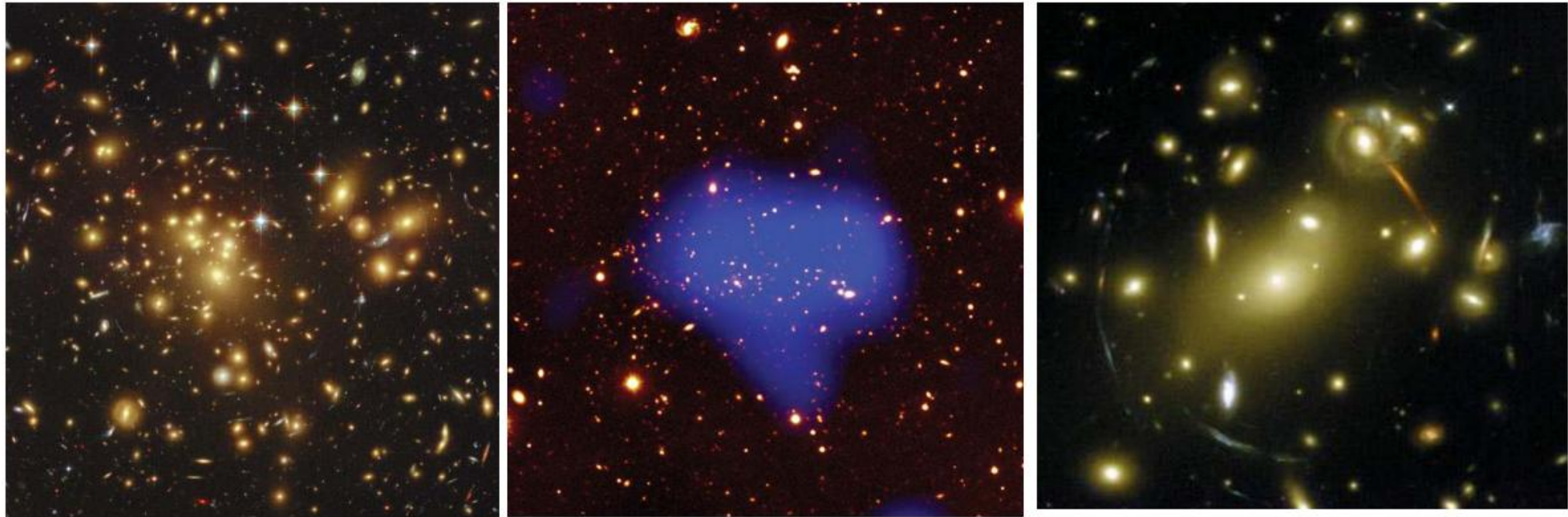
X-rays reveal hot gas between the galaxies.

Measure gravitational lensing



A cluster's gravity bends background light. The distortions reveal its mass distribution.

Compare independent measurements



Galaxy motions, X-ray gas, and lensing all support substantial unseen mass in clusters.

Test the explanation

Additional matter

Dark matter supplies gravitational mass beyond the observed stars and gas.

Test this model across galaxies, clusters, and cosmology.

Modified gravity

Alternative models change the gravitational law.

A successful explanation must address all the observations together.

What could dark matter be?

Compact objects

MACHO means massive compact halo object.

Examples include faint stars and stellar remnants.

New particles

WIMP means weakly interacting massive particle.

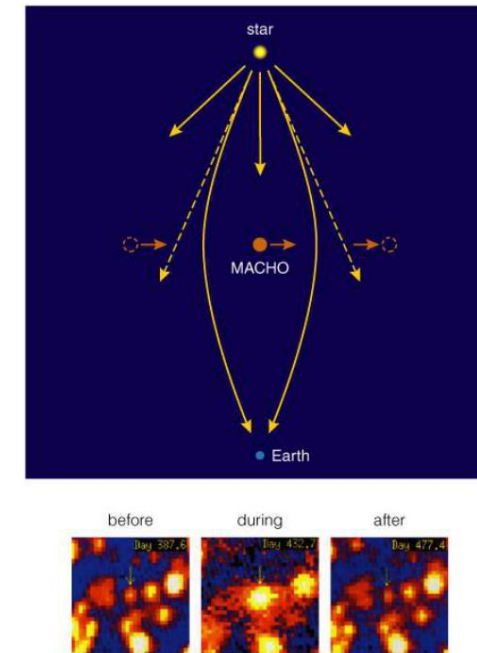
WIMPs are one proposed particle candidate, not a confirmed discovery.

Search through microlensing

Microlensing surveys search for these brightening events.

The results constrain compact objects in Galactic halos.

Ordinary faint objects cannot explain all the dark matter.



A foreground compact object can temporarily magnify a background star.

Particle models remain hypotheses

Why consider particles?

Early-universe models can produce particles that survive today.

Their gravity can affect the growth of galaxies.

What would confirm them?

Search for particle interactions and astrophysical signatures.

A successful model must agree with measured limits.

Dark matter and dark energy

Dark matter

Unseen matter inferred through gravity.

It helps explain the motions and growth of cosmic structures.

Dark energy

The name for whatever drives the accelerated expansion of the universe.

Its physical nature remains uncertain.

The cosmic mass-energy budget

Component	Approximate share
Ordinary matter	5%
Dark matter	27%
Dark energy	68%

Present-day estimates in the standard cosmological model. Source: NASA Space Place.

Discuss the evidence

How do dark matter and dark energy differ?

What do galaxy rotation curves tell us?

Why compare three cluster-mass measurements?

What evidence could distinguish dark matter candidates?

Teaching references

Original lecture

Credited to Mark Voit in the source collection.

Original reading prompt:

LT Book

“Dark Matter,” 143, #1-9

Additional reading

[NASA dark matter overview](#)

[NASA Space Place](#)

Original figure credits and rights remain in effect.